

Office of Chief Engineer (Planning)

Maharashtra State Electricity Distribution Co Ltd
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Ref. No. HO/CE Planning/ 01

Date: 02.01.2022

PROOF OF CONCEPT (PoC) AGREEMENT GIS Survey of Electrical Network Assets for NA Readiness

Parties:

1. **Maharashtra State Electricity Distribution Company Limited (MSEDCL)**
 - o Address: Office of Chief Engineer (Planning), 5th Floor, Prakashgad, Plot No. G-9, Prof. Anant Kanekar Marg, Bandra (East), Mumbai - 400051.
 - o (Hereinafter referred to as "The Utility" or "MSEDCL")
2. **M/s. Nibe Space Private Limited**
 - o Address: M/s NIBE SPACE PRIVATE LIMITED, 202 Windfall, Sahar Plaza, C Wing, Andheri (East), J.B. Nagar, Mumbai- 400059, Maharashtra
 - o Represented By: Dnyaneshwar Nibe, Director
 - o (Hereinafter referred to as "The Service Provider" or "M/s Nibe Space Private Limited")

Scope of PoC:

- The bidder shall conduct a pilot GIS survey of electrical distribution assets (HT/LT line, poles, DTCs, etc.) for an area in Rahuri Sub division (at least 1 complete feeder):
 - Substation: 33/11 KV RAHURI KD,
 - Division: SHRIRAMPUR RURAL DIVISION,
 - Subdivision: RAHURI S/DN.,
 - Pin Code: 413705
- Employ the proposed AI/ML-driven satellite imagery analysis, LiDAR, drone surveys, or street view analytics to capture the entire distribution network, including assets, conductors, and connections.
- Ensure the generated GIS data reflects accurate network topology, with proper connectivity, snapped at zero tolerance, and free from disconnection errors.
- Split all cables and lines at every junction, maintaining correct electrical flow direction, phase, voltage, and circuit relationships.
- Populate asset attributes using AI/ML tools, validated through QA/QC processes.
- Migrate the surveyed Data into MSEDCL GIS & NA systems (ArcGIS and CYMDIST System) as per the requirements of MSEDCL.
- Provide a demonstration of the web portal and mobile app functionalities for network validation, attribute editing, and delta survey updates.

Evaluation Criteria:

- Completeness and accuracy of the GIS network topology and connectivity.
- Demonstration of advanced technology integration and automation capabilities.
- Ability to meet the specified network validation, QA/QC, and attribute population standards.
- Timeliness and quality of deliverables during the PoC period.
- Successful migration into MSEDCL GIS & NA system without manual intervention.

The PoC will be carried out at no cost to MSEDCL

Expected Benefits to MSEDCL:

- State-wide Asset Visibility through complete 33/22/11 kV network mapping enables better planning, monitoring, and maintenance of distribution assets.
- Creation of a Digital Twin provides a scalable and future-ready platform for grid modernization across MSEDCL.
- Identification of Congestion and Equipment Overloads minimizes outages, enhances network reliability, and supports optimized capital expenditure.



- Data-driven Capex Planning and Loss Reduction ensure efficient investment decisions and improved financial performance.

Technical Standards and Data Validation

The data quality and structure are paramount and will be the sole criteria for acceptance.

1 Positional Accuracy Requirements up to 0.5 meter

2 Data Model and Attribute Accuracy

- **Data Model Compliance:** M/s Nibe Space Private Limited shall strictly adhere to the MSEDCL GIS Data Model Schema and all mandatory attribute fields provided by MSEDCL.
- **Attribute correctness:** > 99.5%
- **Photo Validation:** Every major asset must have a geo-tagged photograph confirming its physical existence and nameplate details.

3 Network Topology and Validation (NA Readiness)

The collected data must be structured to form a valid, connected Geometric Network to enable Network Analysis.

1. **Connectivity:** All line features must **snap** precisely to the respective point features (Poles, DTCs, etc.). Zero dangles or undershoots are acceptable.
2. **Source Trace:** A Trace Downstream analysis initiated from the 11kV Feeder switch point in the GIS must successfully reach **every mapped asset** in the sample area, confirming full connectivity.
3. **DTC-LT Circuit Linkage:** The data structure must accurately reflect the number of LT outgoing circuits from the DTC and their respective consumer connections.
4. **Unique Asset IDs:** Every feature must be populated with a **Unique MSEDCL Asset ID** compliant with MSEDCL's current naming convention.

Commercials and Terms

1 No POC Fee and Payment will be claimed by M/s Nibe Space Private Limited.

2 MSEDCL Responsibilities

- Appoint a Nodal Officer for daily coordination and site access facilitation.
- Provide the MSEDCL GIS Data Model Schema and Attribute Domain Values.
- Grant necessary system access for the final data upload and NA demonstration.

3 Intellectual Property

All raw data, intermediate files, processed data, final GIS databases, and reports created during the execution of this POC shall be the sole and exclusive property of MSEDCL. M/s Nibe Space Private Limited shall not retain, use, or disseminate the data for any purpose other than fulfilling this agreement.

Acceptance and Termination

1 Acceptance Criteria

The POC shall be deemed "Successfully Completed and Accepted" only upon the formal written approval from the Chief Engineer (Planning), MSEDCL, following confirmation of the three conditions:

1. **Data Quality Check:** The data meets all accuracy and attribute standards.
2. **GIS Portal Integration:** The final data set is successfully loaded and viewable on the MSEDCL GIS Portal.
3. **NA Readiness Demonstration:** MSEDCL's NA team successfully imports the data and runs a basic Load Flow Analysis (LFA) without topology-related errors.



2 Termination Clause

MSEDCL reserves the absolute right to terminate this POC agreement at any time, with written notice, if the quality, accuracy, or topological readiness of the data falls below the minimum standards specified. In the event of such termination due to non-compliance, M/s Nibe Space Private Limited shall not be entitled to any payment, compensation, or reimbursement whatsoever.

SIGNATURES

MSEDCL (The Utility)

**M/s Nibe Space Private Limited (The
Service Provider)**

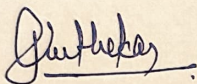
Name:

Name: Dnyaneshwar Nibe

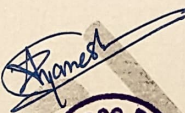
Designation: Chief Engineer (Planning)

Designation: Director

Sign:



Sign:



Seal:

**Chief Engineer (Planning)
MSEDCL**

Seal:

